

BREAKHOOK · DESIGN DECISIONS REGISTER

BreakHook design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/breakhook		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Fork prop at mid-length decided by Amish (6A, BHK-DDR-003); new open decision on the prop holder's distance from the wall; value engineering re-costed
0.1	2026-10-03	Amish Chadha	Register opened at TRL 3; all decisions made under Amish's 2026-10-03 pre-approval

BreakHook design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from.

Safety: Decisions that touch safety (the pole withdrawn before the haul, the 3 m overhead line rule, hauling distance, proof loading, scope limited to free-standing single-storey dwellings) took the conservative option. Each record states what evidence would relax it.

■ Open decisions

#	To be decided	State	Options	Recommendation	Affects in the build	Source
1	Where the prop holder stands while the hook is set	The fork prop works only near the pole's mid-length; there its holder stands about 2.9 m from the wall for the 25 s of placement, 1.1 m nearer than R1's 4 m front-hand distance. The rear handler is 5.06 m away.	(a) Accept 2.9 m for placement only, restating R1's distance as applying to the pole hands, with the prop holder withdrawing with the pole before the haul; (b) keep 4 m for everyone and move the prop back (droop 0.72 m, no better than by hand); (c) keep 4 m and set the prop, then let it stand alone in the fork while its holder steps back (untested)	(a), checked in the TRL 4 reach trial and drill: the firebreak is opened two dwellings ahead of the fire (BHK-DDR-001, decision 10), so the dwelling being hooked is not yet burning, and the prop only helps at mid-length	None to the parts; the check card and drill wording	BHK-DDR-003 ; BHK-CAL-001 [F19], [F24]

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The pole tube carries the maker's dielectric test certificate to ASTM F711 for live-line tool tube, and its stated test voltage	R4 is met only by this certificate	BHK-DDR-001, decision 1
2	Actual outside diameter of the 44.5 mm tube and bore of the 50.8 x 3.2 sleeve tube; how much sanding gives a 0.2 mm slip fit	Nominal sizes give no clearance	BHK-DDR-002
3	Stiffness of the tube from the maker's data (E along the tube); measured hook droop at 25° on the fork prop	The 0.27 m droop estimate with the prop (0.66 m by hand) uses $E = 20 \text{ GPa}$	BHK-CAL-001 [F8], [F16]
9	Outside diameter of the 25.4 x 3.2 prop tube and bore of the 32 x 3 tube; the upper tube slides freely	0.3 mm radial clearance on nominal sizes	BHK-DDR-003
4	Bore of the 50.8 x 2.0 steel socket tube (46.8 mm nominal)	Sets the 1.15 mm clearance and the friction ring wrap	BHK-DDR-002
5	Turns of rubber tape that give a 50 N release	R12	BHK-CAL-001 [G1]
6	Shackle jaw width at least 12 mm and pin no larger than 12 mm	Fits the 10 mm plate and the 13 mm hole	bom/bom.csv line 11
7	Breaking strengths on the rope, leader and cord labels	Factors in BHK-CAL-001 section E	bom/bom.csv lines 12, 13, 15
8	Weighed mass of the pole and hook head	R5 has 0.18 kg of margin	BHK-CAL-001 [F10]

■ Value engineering

Value-engineering target: USD 2,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,170 (USD 830 under the target), including two fork props at USD 36 each. Main cost drivers and savings worth trying:

- Certified, foam-filled pole tube: USD 270 for six sections (25 % of the kit). A bulk order across several communities, or a local pultruder, is the main saving; uncertified tube is not an option (R4).
- Pull ropes: USD 140. Polyester double braid could give way to a cheaper three-strand polyester of the same breaking strength, with a check on hand grip and prusik hold.
- Hook head fabrication: USD 80 of welder time. Plasma-cut plates in a batch of ten would roughly halve it.
- Gloves, proof loading and the rack are each about USD 50 to 60 and are kept for safety and governance.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	Pole: foam-filled, electrical-grade fibreglass tube 44.5 x 3.2 with an ASTM F711 certificate, three 1.8 m sections	Amish Chadha, pre-approval: "I pre-approve the batch runs along with any recommendations you come up with."	BHK-DDR-001, decision 1
2026-10-03	The pole is withdrawn before the haul; the rope alone pulls	As above	BHK-DDR-001, decision 2
2026-10-03	Hook head cut from 10 mm plate, welded in a slotted socket; WLL 3 kN, proof 6 kN before issue	As above	BHK-DDR-001, decision 3
2026-10-03	Wire rope leader, rated shackles, 25 m polyester pull rope	As above	BHK-DDR-001, decision 4
2026-10-03	Crew of four to deploy and about ten haulers on toggles; no pulley in the base kit	As above	BHK-DDR-001, decision 5
2026-10-03	Nearest hauler at 1.5 x the dwelling height and never closer than 10 m	As above	BHK-DDR-001, decision 6
2026-10-03	Never raise the pole within 3 m of an overhead or informal line	As above	BHK-DDR-001, decision 7
2026-10-03	Hand band and 3.0 m minimum insulated length	As above	BHK-DDR-001, decision 8
2026-10-03	Kit governance: locked, sealed rack, two keyholders, use log, pulls only during a fire under the fire plan	As above	BHK-DDR-001, decision 9
2026-10-03	Firebreak at least two dwellings ahead of the fire	As above	BHK-DDR-001, decision 10
2026-10-03	Free-standing dwellings and end units only	As above	BHK-DDR-001, decision 11
2026-10-03	Monthly check card; fresh proof load after a seal break	As above	BHK-DDR-001, decision 12
2026-10-03	First co-design candidate to approach: City of Cape Town Fire and Rescue Service reservists (Western Cape), then Kenya Red Cross Society (not agreed)	As above	BHK-DDR-001, decision 13
2026-10-03	Requirements updated, R11 to R14 added	As above	

Date	Decision	Decided by	Record
			BHK-DDR-001, decision 14
2026-10-03	budget_usd stays at USD 2,000 as the value-engineering target	Amish Chadha: "I also accept any cost overruns or variations from the assumed scope cost."	BHK-DDR-001, decision 15
2026-10-03	Design for construction: plate hook in slotted socket, slip-fit pole with friction ring, bonded sleeves and nylon pins, rope tab with shackle and leader, S355 plate with 36 mm shank and 48 mm arm, prusik toggles, welded wall rack	Amish Chadha, pre-approval as above	BHK-DDR-002
2026-10-03	R1: add a light fork prop that supports the pole at mid-length, carried separately so the pole and hook head stay under 8 kg (decision 6A)	Amish Chadha: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"	BHK-DDR-003